# Hereditary Diffuse Gastric Cancer

## Hereditary Diffuse Gastric Cancer

Genetic syndrome first reported among Maori population in New Zealand

- Autosomal dominant
- Early-onset diffuse gastric cancer ($\tilde{x}$ age = 40)
- High penetrance

## CDH1 mutations and HDGC

Pathologic variants in CDH1 gene associated with HDGC

## Penetrance in Populations Enriched in HDGC

## Prophylactic Gastrectomy

- Stanford

90% of patients undergoing prophylactic gastrectomy harbor microscopic foci of gastric cancer in their stomach

## Endoscopic Screening - NIH

A large NIH cohort study of 270 CDH1 carriers undergoing surveillance with systematic gastric mapping found only two patients developed clinically evident pT3N0 gastric carcinomas over a median 31-month follow-up, with no gastric cancer deaths.

(Asif et al. 2023)

## Lifetime Risk of Cancer

| Cancer endpoint | Cumulative risk to age 80 | Mean age at onset | References |
|---|---|---|---|
| **Advanced gastric ca ( stage II), males** | 10.3% | 47-49y | 4][35][2] |
| **Advanced gastric ca ( stage II), females** | **~6.5–7%** | 47-49 | |
| Gastric ca including occult pT1a, males | 37-42% | | 4 35 2 |
| Gastric ca including occult pT1a, females | 25-33% | | |
| **Lobular breast cancer (females)** | **37–55%** | 51–54y | 1 2 3 8 |

## Endoscopic Screening - Cambridge

imilarly, a 16-year longitudinal cohort study demonstrated that endoscopic surveillance with systematic random biopsies can identify early lesions and inform decision-making about gastrectomy.

(Lee et al. 2023)

## Endoscopic Screening

2025 study of 20 carriers undergoing long-term surveillance (mean 62 months) found no patients developed aggressive

(greater than pT1a) cancer, even among those with positive pT1a biopsies who continued surveillance.

aside: Expert Endoscopic Surveillance in CDH1 Pathogenic Variant Carriers Seems Safe, Even After Positive (pT1a) Biopsies. Journal of Medical Genetics. 2025. Benusiglio PR, Leenhardt R, Duros C, et al.New

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